Tradeoffs between carbon assimilation and release reveal a key role of the tropics in long-term global-scale carbon sequestration

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The quantification of terrestrial ecosystems' carbon sequestration is crucial for understanding their role in mitigating global warming, with proper assessment requiring consideration of temporal scale. Our study quantified carbon sequestration by integrating carbon inputs and outputs over time to analyze the trade-off between high carbon assimilation rates associated with the tropics and low carbon losses through respiration related to high latitudes, while also assessing the effect of disturbance-related carbon losses. For the analyses, we utilized results from 9 CMIP6 Earth System Models. Then we compared the findings with the results of non-coupled models (TRENDY) and in-situ measured data-based products (CarboScope, FLUXCOM-X, CAMS). Results show that when only assimilation and respiration are considered, tropics exhibit higher carbon sequestration during 1850-2014, but the inclusion of disturbances shifts higher sequestration to high latitudes. Comparisons among products in 2000-2014 reveal similar trends between CMIP6 and FLUXCOM-X for carbon sequestration without disturbances, albeit with lower TRENDY magnitudes. When disturbances are considered, most products indicate sequestration amounts in high latitudes, except CAMS, which favors mid-latitudes. These findings underscore the significant impact of disturbances on terrestrial carbon cycles over the past century, suggesting that ecosystems, particularly in the tropics, had the potential to sequester more carbon than they did.

Global warming mitigation | Terrestrial ecosystems | CMIP6 | Carbon cycle

The capacity of the terrestrial biosphere to take up and store about a third of anthropogenic carbon emissions derived from the combustion of fossil fuels (1) depends to a large degree on the long-term balance between its carbon inputs (assimilation) and outputs (release), which fluctuate over time and space due to natural processes and human activities (2). Carbon inputs to terrestrial ecosystems are controlled primarily by photosynthesis, while carbon outputs arise from respiration of vegetation and soils as well as disturbances such as wildfires and land-use changes (3).

Although there is good evidence that the terrestrial biosphere has acted as a significant sink of carbon during the previous decades (1, 4), there is large uncertainty regarding the stability and persistence of this sink (5). For instance, measurements of atmospheric radiocarbon have shown that the terrestrial biosphere switched from a sink to a source of radiocarbon produced by thermonuclear-weapon testing about two decades after its spike in the atmosphere (6, 7). Similarly, recent measurements of radiocarbon in ecosystem respiration have shown a one- to two-decade time lag between photosynthesis and respiration (8, 9). These measurements indicate that carbon uptake from the terrestrial biosphere is temporary, and the large majority of carbon fixed by photosynthesis returns to the atmosphere in a matter of years to a few decades (10). Therefore, the large land carbon sink on land may soon be overtaken by a lagged respiratory response that could turn the land biosphere into a net carbon source.

A key metric to understand the temporary nature of the balance between carbon assimilation and release, as well as the lag times between carbon sinks and sources, is the transit time of carbon. This metric, however, is very difficult to determine for the nonlinear and non-equilibrium conditions of the coupled carbon-climate system (11, 12). Recent studies have shown that, for long time periods, it is possible to obtain the cumulative lag effects between assimilation and release, implicitly defined by transit times, without explicitly computing this metric (13). Instead, an approach that relies on the mathematical integration (double integral) between instantaneous sources and sinks can be employed. This approach, which we define here as integrated Carbon Sequestration (CS), can be applied to the large volumes

Significance Statement

Understanding terrestrial ecosystems' role in carbon sequestration is crucial for addressing global warming. This study provides key insights by analyzing carbon dynamics across latitudes and time scales, incorporating both natural processes and human-induced disturbances. By comparing various models and data products, we reveal that tropical regions show higher carbon sequestration potential when considering only assimilation and respiration. However, including disturbances shifts this advantage to higher latitudes. These findings highlight the significant impact of disturbances on global carbon cycles over the past century and suggest untapped sequestration potential, particularly in tropical ecosystems. Our research emphasizes the importance of comprehensive, long-term assessments in developing effective climate mitigation strategies and underscores the need for targeted conservation efforts to maximize the carbon sequestration capacity of terrestrial ecosystems worldwide.

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of available information on net carbon fluxes between the land and the atmosphere derived from both observational products and models.

Our aim in obtaining this metric from existing information is to learn from the historical record on the trade-off between carbon sinks and lagged carbon release. For example, although tropical forests have large photosynthetic fluxes, carbon returns back to the atmosphere relatively quickly (9, 14). On the contrary, high-latitude ecosystems with lower photosynthetic rates return carbon to the atmosphere more slowly than tropical ecosystems (15, 16). Therefore, we ask here the question, does the tradeoff between carbon assimilation and transit time favors carbon sequestration in regions with large assimilation rates but fast transit times such as the tropics, or regions with low assimilation rates and slow carbon release such as high-latitude regions? In addition we ask, to what degree disturbances related to fires and land-use change modify not only the amount of carbon uptake but also the amount of time that carbon can be retained in terrestrial ecosystems?

We calculated CS for the period 1850–2014 using monthly outputs from gross primary productivity (GPP), heterotrophic respiration (r_h), autotrophic respiration (r_a), and net biome production (NBP), based on output from nine Earth System Models (ESMs) participating in the Coupled Model Intercomparison Project Phase 6 (CMIP6). We also used numerical output from land surface models participating in the TRENDY project. To contrast modeling results with observations, we calculated CS for a shorter period (2001–2014) using results from two global atmospheric inversions that assimilate data on mole fraction of CO_2 and provide gridded net exchange fluxes of carbon between the land and the atmosphere (CarboScope, and CAMS), and a product that assimilates data on net ecosystem exchange, land surface cover, and meteorology (FLUXCOM-X).[*]

By accurately quantifying terrestrial carbon sequestration, we aim to deepen our understanding of ecosystem carbon dynamics to inform better land management practices and enhance strategies to maximize ecosystems' capacity to mitigate climate change effectively.

Results

Spatial patterns of Carbon Sequestration integrated over 165 years. Carbon Sequestration (CS), obtained as the double integral of NEP from eight CMIP6 models (Supplementary Table 1 (see SI.tex)), reveals a predominantly positive global distribution, with notable exceptions in polar regions and hyper-arid zones (Fig. 1a).[†] Forests exhibit the highest CS values, particularly in the Amazon basin, Eastern and Western North America, the Congo basin, Western Europe, and South East Asia. The latitudinal CS average shown in Fig. 1b demonstrates positive CS across all latitudes, indicating long-term carbon sequestration globally during the entire 165 year simulation period (1850-2014). Peak CS occurs in tropical latitudes (approximately 25°S to 20°N), while mid-latitudes (approximately 20-40°N and 20-30°S) show the lower values. Higher values of CS in tropical and boreal regions in comparison to mid-latitudes suggests that high C assimilation or slow C release contribute to retain carbon in the long-term, but the effect of high C assimilation is stronger as suggested by the high CS values in the tropics.

Integrated CS across all latitudinal zones exhibits a consistent upward trend over time in all models, albeit with varying magnitudes. The NOAA-GFDL-ESM4 and ACCESS-ESM1-5 models show the highest CS values, while the CanESM5 and BCC-CSM2-MR models show the lowest values (Fig. 2a). All models predict that tropical regions retained the largest amount of carbon×time from 1850 to 2014, followed by mid-latitudes (Fig. 2b-d). Although high latitudes sequester on average more carbon×time per unit area than mid-latitudes (Fig. 1b), the larger territorial expanse of mid-latitudes results in a greater total CS (Fig. 2). All CMIP6 models indicate a positive CS in the tropics during the studied period (Fig. 2b). In mid- and high-latitudes, at least one model (BCC-CSM2-MR) shows negative CS, where the accumulated balance between respiration and GPP results in a net loss over the 165-year period (Fig. 2c,d). Additionally, three models indicate nearly zero CS in high latitudes (Fig. 2d).

Disturbance impacts on Carbon Sequestration integrated over 165 years. The previous results, based on the double integral of NEP, only account for differences between GPP and ecosystem respiration and exclude fluxes due to disturbances like deforestation and fires. When using the double integral of Net Biome Production (NBP) to obtain CS, spatial and temporal dynamics change significantly, with values about two orders of magnitude smaller than those obtained with NEP (Figs. 2a and 3a). These values are predominantly negative, which indicates that over the entire 165-year timeframe the accumulated carbon losses are higher than carbon gains producing a net carbon debt. Globally accumulated CS is negative for all CMIP6 models except CNRM-ESM2-1 and ACCESS-ESM1-5, with the ensemble indicating a carbon debt of approximately 0.8 PgC yr for the period 1850-2014 (Fig. 3a). High-latitude regions show the least negative values, followed by the tropics (Fig. 3b-d). Negative CS values are observed in 5 out of 8 models for the tropics, 6 for mid-latitudes, and 5 for high latitudes (Fig. 3b-d). Notably, the ensemble accumulation of CS over time is no longer monotonically increasing as when using NEP; after decreasing from 1850 to around 1970, CS tends to slightly increase showing a recovery trajectory.

The difference in CS calculated with NEP versus NBP is dominated by differences occurring in tropical latitudes. Globally, the difference between CS for these two type of fluxes is 2.4×10^{21} PgC yr, with tropical latitudes accounting for 58% of this difference (1.4×10^{21} PgC yr). Therefore, fluxes not accounted for in NEP but included in NBP have a strong long-term effect in carbon sequestration, and they are particularly important in the tropics.

Comparison with observation-based products. For the 14-year period when model output and observation-based products overlap (2001–2014), integrated CS consistently show higher values when derived from NEP compared to NBP (left and right panel of Fig. 4, respectively). NEP-based CS from CMIP6, TRENDY, and FLUXCOM-X indicate that tropical regions have the highest values, while high latitudes

[*] I think it's better to provide the citations of these products than the links. The links won't work in a few years, and the authors usually prefer a citation.

[†] Figure 1 can be improved by moving the legend to the right so it doesn't overlap with the the results for low latitudes. The labels a and b do not need a parenthesis, you can remove them.

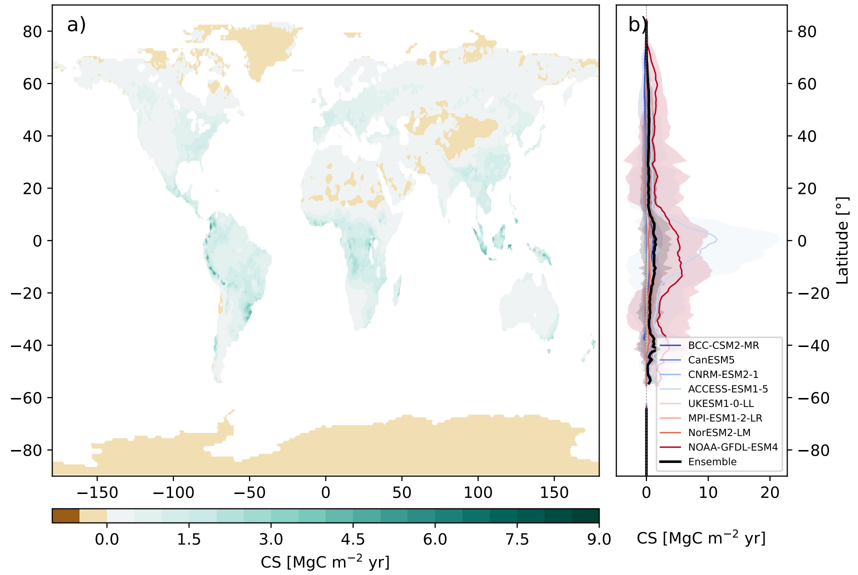

Fig. 1. Carbon Sequestration (CS) spatial patterns from NEP from CMIP6 models (1850-2014). a) Global distributions of CS based on the weighted ensemble of 8 CMIP6 models, and b) latitudinal CS profiles for individual CMIP6 models (colored lines) and their ensemble (black line). The dotted line in b) represents null values of CS and the shadow areas indicate two standard deviations from the mean.

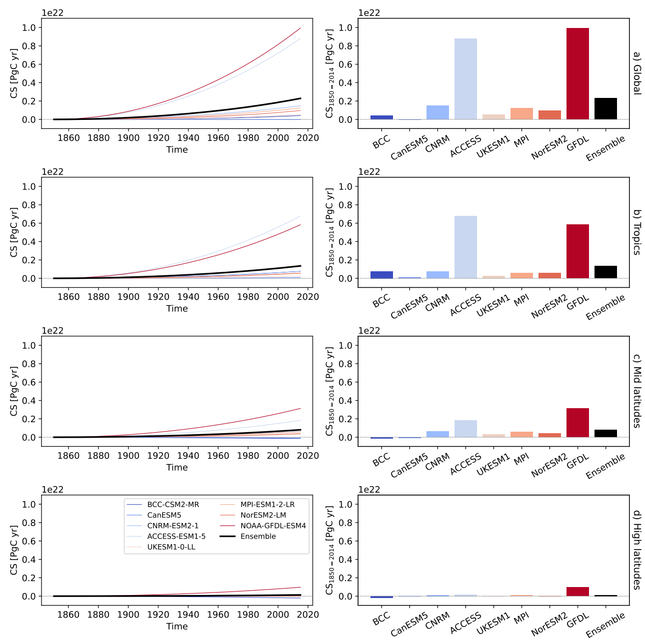

Fig. 2. Time accumulation of Carbon Sequestration (CS) (left panel) and CS at the end of the period from 1850 to 2014 (right panel) calculated using NEP for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes zones calculated using the results of 8 models from CMIP6. Black lines and bars indicate the results using the ensemble of the models. The dotted lines indicate null values of CS. To save space, only the initial part of each model name is displayed on the horizontal axes in the right panel.

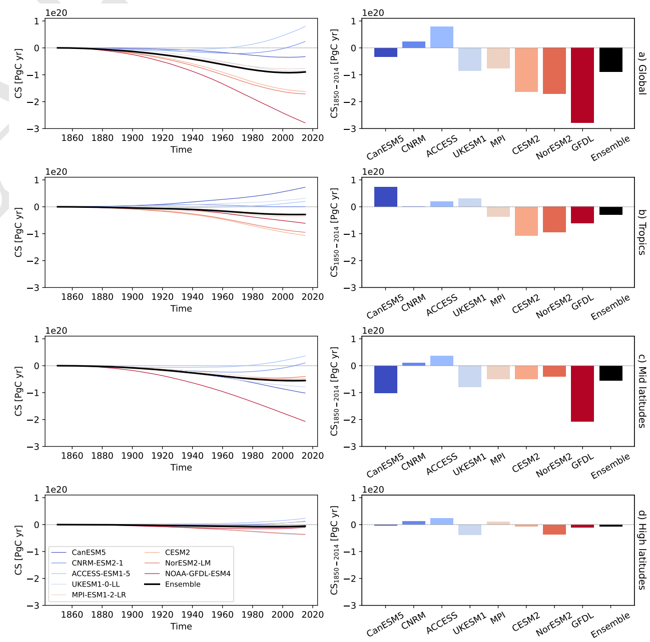

Fig. 3. Time accumulation of Carbon Sequestration (CS) (left panel) and CS at the end of the period from 1850 to 2014 (right panel) calculated using NBP for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes zones calculated using the results of 8 models from CMIP6. Black lines and bars indicate the results using the ensemble of the models. The dotted lines indicate null values of CS. To save space, only the initial part of each model name is displayed on the horizontal axes in the right panel.

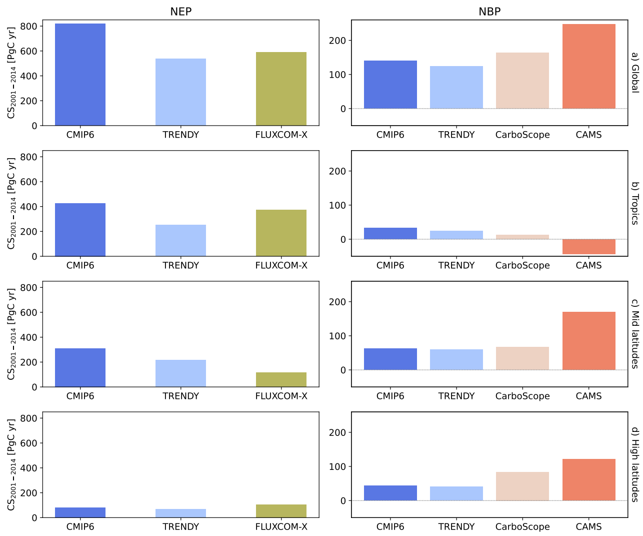

Fig. 4. Carbon sequestration (CS) across different products (2001-2014) calculated using NEP or NEE (left panel) and NBP (right panel) for a) global, b) tropics, c) mid-latitudes, and d) high-latitudes. The dotted lines indicate null values of CS. CMIP6 and TRENDY CS values correspond to the weighed ensemble of their respective models. Note that the y-axis of the left and right panels do not have the same scale.

have the lowest, aligning with previous results in Fig. 2 for this period. The observation-based product FLUXCOM-X provides values of CS in between CMIP6 and TRENDY at the global level, with lower values at mid latitudes suggesting a slight overestimation of CS by the model ensambles. Although both ensemble models display similar latitudinal zone trends, CMIP6 consistently exceeds TRENDY by up to 250 PgC yr globally.

When using NBP, CS is predominantly positive for all products except CAMS in tropical regions. In fact, the CAMS observational product produces the higher differences in CS in comparison to the CarboScope product and the two model ensambles. The CarboScope product suggests overestimation of CS by the two model ensambles in tropical latitudes and overestimations in high-latitude regions (Fig. 4).

The highest values of NBP-based CS were obtained in mid-latitude regions (right panel in Fig. 4). The shift from negative CS values obtained with NBP in the longer 165-year period to positive values in the shorter 14-year period indicates that legacy effects of previous disturbances are not taken into account for the 2001-2014 period. This is particularly noticeable in the mid-latitude regions where carbon recovery since the mid 20th century is the dominant process taken into account in this short calculation period, and previous disturbances are ignored and play no role in the computation of CS.

In the tropics, the change from NEP-based CS to NBP-based CS is large for the CMIP6 and TRENDY models during this 14-year timeframe. It changes from 427 PgC yr to 33 PgC yr (394 PgC yr difference) for CMIP6 models, and from 252 PgC yr to 24 PgC yr in TRENDY (228 PgC yr difference). This difference is much larger than for other latitudes and it is responsible for the larger differences observed at the global level between CS based on NEP versus NBP (Figure 4).

Discussion

Trade-off between high carbon assimilation and lagged respiratory responses. Our approach of computing Carbon Sequestration (CS) as the double integral of carbon exchange between terrestrial ecosystems and the atmosphere provides a method to evaluate tradeoffs between carbon assimilation and lagged respiration responses over long timescales. This approach implicitly takes into consideration the transit time of carbon through terrestrial ecosystems and quantifies cumulative effects of delayed release of carbon fluxes from ecosystems back to the atmosphere. Using this approach, we found that carbon sequestration can be particularly high due to high assimilation rates as in tropical latitudes or due to slow respiration responses as in high latitudes, but the assimilation effect dominates globally over the respiration effect. Therefore, the tropics emerged in our analysis as the region where more carbon can be retained longer over decadal to centennial timescales in the absence of disturbances.

The monotonic increasing trend of NEP-based CS for all CMIP6 models, is mostly the effect of larger GPP than Re fluxes for all regions during the entire simulation timeframe.[‡] Independently on whether this is an artifact of models or a real sustained carbon sink due to CO_2 and N fertilization, increased water use efficiency, or prolonged growing season (1, 17–20), the key insight from our results is that as long as GPP monotonically increases, the lagged respiration response of the biosphere does not catch up and large quantities of carbon×time are retained out of the atmosphere on land. However, this sustained monotonic increase in NEP-based CS is only hypothetical because it does not include other important fluxes such as fires and land-use changes that completely change CS trajectories.

Consequences of disturbances. Factors beyond climate such as deforestation and fires significantly decrease values of CS. From 1850 to 2014, most models indicate that cumulative carbon release exceeded assimilation (Fig. 3), putting terrestrial ecosystems in a net carbon×time debt with respect to pre-industrial carbon stocks. While the Amazon and Congo basins as well as Papua New Guinean forests still exhibit high positive CS values at the end of the simulation period (Supplementary Fig. 5)[§], mid- and high-latitude regions show large negative CS values that lead to a large global-scale carbon×time debt (Fig. 3). This debt increased over time since the beginning of the industrial revolution in the global ensamble (Fig. 3) driven mostly by large release fluxes in the tropics and mid-latitude regions. After the 1970s a recovery trajectory in CS indicates that the debt has been partially paid back by forest recovery in mid-latitudes, but the size of the carbon×time debt is still very large at the end of the simulation period, particularly in Easter North America, Central America, Northern Andes and Eastern South America (Supplementary Fig. 5).

Differences between NEP- versus NBP-based CS indicate that these disturbances play a very large role in the ability of the biosphere to fix carbon out of the atmosphere and retain it for long periods of time. These differences are particularly large for the tropics where carbon fixation is large, but release

‡ I think we need to include in the SI the plots of GPP and Re for all models and the ensamble. This is important to be able to explain the results.

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fluxes from fires and disturbances hinder the ability of these regions to retain carbon over decadal timescales. Empirical studies have found that the mean age of respired carbon in tropical forests is on the order of one to two decades (9, 14), which implies that lagged respiration effects in the tropics are intrinsically fast, and can be exacerbated by additional losses due to fires and deforestation that would drastically reduce transit times in these regions. Therefore, when evaluated with the CS metric, the loss of carbon×time in the tropics is much larger than in other latitudinal regions.

When only data from 2001-2014 is considered, CS calculated with NBP becomes positive, albeit much lower than CS calculated with NEP for the same period (Fig. 4). This shift in trend demonstrates how carbon flux balances can vary significantly depending on the chosen time horizon. It shows that over these 14 years, carbon inputs can offset recent losses. However, these short-term gains are insufficient to counteract the long-term accumulated carbon debts. These gains could also be noted in the change of slope observed from around the 1970s for NBP-based CS (left panel in Fig. 3a).

Agreement and discrepancies among products. [¶]

Implications for climate change mitigation. Despite renewable energy growth and some decarbonization, fossil fuels still dominate to meet rising global energy demand, causing emissions to increase and hindering Paris Agreement goals (20, 21). While terrestrial ecosystems help mitigate these emissions (17, 20, 22), some regions are carbon neutral or sources, and future projections suggest ecosystems may become net carbon sources overall (20, 23). Our results show that terrestrial ecosystems sequestered carbon across all latitudes when considering only biospheric fluxes of assimilation and respiration over the 165 years leading up to 2014 (Fig. 1). However, disturbances can shift this balance negatively (Fig. 3a). Interestingly, even when considering disturbances, CS was positive globally from 2001-2014, except in the tropics according to CAMS (right panel in Fig. 4). These findings underscore how the chosen time horizon can significantly influence whether an ecosystem appears as a carbon source or sink.[‖] This could be explained by the various processes operating on different time scales, which may not be fully captured in short-term carbon flux measurements (24, 25). Therefore, when evaluating carbon dioxide removal, capture, and storage strategies, it's crucial to quantify carbon sequestration by considering not only how much carbon is in the ecosystem, but how long it remains there (26, 27).

Past disturbances such as deforestation, fires, and land-use changes have significantly impacted carbon sequestration by terrestrial ecosystems, particularly in the tropics. The substantial difference between CS calculated with NEP and NBP underscores the high potential of terrestrial ecosystems to sequester carbon. This is particularly noted in the tropics, emphasizing the urgent need for preservation and sustainable management of tropical ecosystems as a priority in anthropogenic CO_2 mitigation.

While this work provides clear insights, significant discrepancies exist between the products used, particularly in the tropics and high latitudes (20). Earth System Models and Dynamic Global Vegetation Models are crucial for projecting future climate scenarios (1) but still struggle to fully capture complex natural interactions (1, 18, 28, 29). This highlights the need for continuous improvement in model development and refinement, particularly regarding carbon ecosystem residence time (24, 30–33). Considering that multi-model ensembles of intercomparison products have been found to outperform individual models (34, 35), the weighted ensemble calculated for TRENDY and CMIP6 in this study could potentially reduce model bias.

In conclusion, this work emphasizes the importance of considering time when assessing carbon sequestration. Furthermore, the spatial patterns of carbon sequestration, both with and without disturbances, can inform the prioritization of areas for targeted climate change mitigation efforts through carbon sequestration strategies.

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[‖] I think we should stay away from the carbon source/sink terminology. We are doing something different with CS, which is more about carbon retention versus debt. Sources and sinks are the more instantaneous fluxes expressed in units of mass/time, but we are accumulating these fluxes twice, therefore we shouldn't be talking about sources and sinks. Also, most previous papers have already talked about sources and sinks, and we need to highlight that we are not doing that here, we are doing something new!

Materials and Methods

Datasets. We calculated the carbon sequestration using Net Ecosystem Production (NEP) (or Net Ecosystem Exchange (NEE)) and Net Biome Production (NBP) outputs from two model intercomparison products and three data-driven products.

Model intercomparison products. We used the results of Gross Primary Production (GPP), autotrophic respiration (r_a), heterotrophic respiration (r_h), and NBP from a set of Earth System Models (ESMs) participating in the most recent climate-carbon simulations under the sixth phase of the Coupled Model Intercomparison Project (CMIP6) (36). Eight models provided monthly outputs for NBP, while eight (seven overlapping) supplied data for the remaining variables of interest in the esm-hist simulation (see Supplementary Table S1). These simulations span the period from 1850 to 2014 and are forced by prescribed anthropogenic CO_2 emissions rather than CO_2 concentrations. The data were obtained from The German Climate Computing Centre DKRZ. In addition, we used results from Dynamic Global Vegetation Models (DGVMs) compiled by the TRENDY v13 project (37). Twenty models provided monthly outputs of GPP, r_a, r_h, and 18 models provided monthly NBP (see Supplementary Table S2). We used experiment S3, which accounts for variations in atmospheric CO_2 and climate and spans the period from 1700 to 2023. Data were requested through the Global Carbon Budget.

Both CMIP6 and TRENDY intercomparison models were included in the analysis due to their distinct focuses. TRENDY DGVMs focus on simulating terrestrial vegetation dynamics and ecosystem processes, while CMIP6 ESMs provide a representation of the entire Earth system, incorporating simplified versions of DGVMs as subcomponents within a broader framework that also accounts for atmospheric, oceanic, and cryospheric processes (38, 39). For both CMIP6 and TRENDY models, NEP values are calculated as $NEP = GPP - (r_a + r_h)$.

Data-driven products. Besides the model intercomparison products, we utilized data from two atmospheric CO_2 inversion products: CarboScope and CAMS. These products estimate land-atmosphere net carbon exchange fluxes by combining CO_2 concentration measurements, transport models, and information on anthropogenic carbon emission and land-atmosphere carbon exchange (40). Specifically, we used the nbetEXToc_v2024E version of CarboScope (41), which provides daily data from 1957 to 2023 based on measurements from 173 sites. For CAMS (42), we used version v24, which includes monthly data from 1979 to 2020 derived from over 100 surface measurements sites. Additionally, we incorporate Net Ecosystem Exchange (NEE) from X-BASE produced with FLUXCOM-X (43), a machine-learning-based product trained on in-situ eddy covariance data. FLUXCOM-X provides monthly NEE estimates for the period 2001–2021

In this study, we assume that NEE and NEP are interchangeable since both represent the same underlying processes—GPP minus respiration—despite being calculated using different data sources. Note the NEE is negative when the ecosystem assimilates more CO_2 than it releases, which is opposite to the convention for NEP.

Homogenization and ensemble calculations. To enable comparison, data from all products were regridded to a uniform $1^\circ \times 1^\circ$ spatial resolution using the bilinear interpolation method implemented via xESMF (44). We standardized the temporal resolution to monthly for all datasets that were not originally in monthly format. CMIP6 calculations utilized the full-time series (1850–2014), while comparisons across products were restricted to their overlapping period, i.e., 2001–2014. Positive NEP and NBP values were defined as fluxes moving from the atmosphere to the land, which contrasts with the conventions used in inverse models.

Then, to calculate the ensemble of CMIP6 and TRENDY models, we used a weighted averaging approach rather than a simple equal-weight average. This procedure assigns weights to each model based on its predictive performance against available data, following the multimodel inference framework presented by Burnham & Anderson (45), which builds on concepts developed by H. Akaike in the 1970s. The weighting procedure calculates each model's weight by measuring its distance from observed data. However, since we are dealing with global spatial models where direct observations are not available, we used data-driven products derived from sparse observations and scaled to the terrestrial surface. Specifically, we used FLUXCOM-X to calculate the NEP ensembles and CarboScope for NBP ensembles (chosen for its longer time series compared to CAMS). The calculation of weights (w_i) for each model i is detailed in (cite text with method). Once w_i are determined, the model ensemble ($\overline{x}$) is computed as

$$\overline{x} = \sum_{i=1}^{k} w_i \cdot x_i \qquad [1]$$

where k is the total number of models, and x_i are the values of the variable of interest for model i. In this case, x corresponds to NEP or NBP. The same weighting procedure is applied to create the ensemble of models from CMIP6 and TRENDY, using all data from periods that overlap with the available data periods of CarboScope (CMIP6:1957-2014, TRENDY:1957-2023) or FLUXCOM-X (CMIP6:2001-2014, TRENDY:2001-2021) (see Supplementary Figs. 1-4).

Carbon sequestration. After harmonizing all NEP (or NEE) and NBP data to the same temporal and spatial scale and calculating the ensembles for CMIP6 and TRENDY, integrated Carbon Sequestration (CS) is computed. Here, CS is defined as the storage of a certain mass of carbon (M) within a system over a given period ($t_0 + T$). It is quantified as the area under the curve representing the evolution of carbon mass over time (26), as

$$\mathrm{CS}\,(t_0, T) := \int_{t_0}^{t_0+T} \int_{t_0}^{t} s\,(t) - r\,(t)\, dt dt. \qquad [2]$$

where $s(t)$ is the gross flux rate of carbon removed from the atmosphere and stored into a system at a given time t, while $r(t)$ is the gross flux rate of carbon released from the system back to the atmosphere. This definition of CS accounts for both the fate of new carbon inputs and the legacy carbon already present in the system at t_0 (26). It considers the total carbon mass under evaluation as the net balance between all inputs and outputs over the specified time horizon (see Fig. 1 in Muñoz et al. (27)).

When calculating global carbon sequestration or aggregating it by latitudinal zones, the grid areas are factored into the computation, removing spatial units. Latitudinal zones are defined as tropical: 22.5°S-22.5°N, mid-latitudes: 22.5°S-55.0°S and 22.5°N-55.0S°N, and high latitudes: >55.0°N.

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